Bawsla

The AI of the User module

Machine Learning + LLM, explained page by page and button by button

Mini-project ESPRIT 5SAE9 · IA for Software Engineering · 2026-2027

Module User · October 2026

1. Two kinds of AI that work together

- **Machine Learning (the brain that calculates).** Three models that learn from data, then predict. They work with **numbers only**.
- **The LLM, Gemini (the brain that writes).** It takes the results of the models and puts them into sentences a person can understand and act on.

Golden rule: the Machine Learning decides, the LLM explains. The LLM never invents a result: it rewrites what the models calculated.

2. Where the data comes from

Before any AI works, the module gathers **11 numbers** about each learner:

Source	Numbers
Quiz module	quiz average, share of correct answers, number of quizzes taken, missed notions (Jointures, Index…), number of mistakes
Cours module	courses followed, courses finished, progress (chapters done), time spent reading (counted minute by minute)
User module	active days in the last 30 days, days since the last login

The AI only receives these numbers, plus course titles and notion names. **Never** a name, an email or an identifier. That is why the Progrès page says « l'IA ne reçoit que des chiffres ».

3. The 3 Machine Learning models

3.1 K-Means: the learning profile

- **What it does:** groups learners who behave alike.
- **How it learned:** it looked at 200 simulated learners and found 3 natural groups by itself, without being told the answer (unsupervised learning).
- **The 3 groups: Régulier** (good pace, correct results), **Débutant en difficulté** (active but many mistakes), **Décrocheur** (little activity lately).
- **For a new learner:** it compares their 11 numbers with the 3 groups and picks the closest.
- **Example:** Lina (20 %, 1 chapter, 2 min) → Décrocheur. Sami (100 %, 3 chapters) → Régulier.

3.2 Random Forest: the failure risk

- **What it does:** estimates the probability that a learner will fail.
- **How it learned:** it saw 200 simulated learners **and** whether each one passed or failed (supervised learning). It is a « forest » of many small decision trees that vote together.
- **What it gives:** a percentage, a level (Suivi normal / À surveiller / Risque d'échec) and the **3 main reasons** (for example « Note moyenne basse aux quiz (20%) »).
- **Who sees it:** only the trainer and the admin, **never the learner** (to avoid discouraging them).
- **Example:** Lina 72 % (high); Sami 58 % (to watch: only 1 day and 5 minutes of activity).

3.3 Collaborative filtering: course recommendations

- **What it does:** suggests courses.
- **How it works:** « learners similar to you followed these courses », combined with the learner's interests and missed notions. It is a hybrid: 50 % similar learners, 30 % declared interests, 20 % popularity.
- **Rules:** never a course the learner already follows; each suggestion comes with its reason.

- **Example:** Lina likes Informatique and follows « SQL pour débuter » → « SQL avancé » is recommended.

4. Page by page, button by button

4.1 Learner → Progrès

What you see	What happens behind
The 4 numbers at the top (active days, quiz average, courses followed, notions to review)	No AI: the learner's real data, collected from Quiz and Cours.
« Cours recommandés pour toi » + Voir	**Collaborative filtering** chooses the courses; « Voir » opens the course.
« Ton profil d'apprentissage »	**K-Means** gives the group and its description.
« Mettre à jour mes centres d'intérêt »	Opens Mon parcours: changing the interests changes the recommendations.
« Ce que ta boussole prend en compte »	Shows which data is available (« Pris en compte »). If data is missing, the AI says so and does not guess.
« Obtenir mon conseil » (LLM)	The module sends Gemini the learner's numbers, profile, missed notions and recommended courses. Gemini writes **2–3 encouraging sentences and 1 to 3 actions**, in « tu ». If the text mentions risk, failure or dropping out, the server **rejects** it.
« Préparer mon plan » (LLM)	Gemini builds a **plan for the next 7 days** (one session per day, with a duration) from the courses in progress, the missed notions and the time chosen in the parcours. The server checks that the days are valid and that the total **does not exceed** the declared time (« 1 à 3 h » = 3 h maximum).
« Régénérer » (tip or plan)	If the data **has not changed**: the **same saved text**, without calling Gemini again (saves the free quota). If the data changed (new chapter, new quiz): a new text.

4.2 Learner → Mon compte → Mon parcours

What you see	What happens behind
The 4 lists (profile, goal, domains, weekly time)	No AI: what the learner declares. Domains feed the recommendations; the time feeds the plan.
« Suggérer mes domaines » (LLM)	The learner writes one sentence (« je veux devenir data analyst »). Emails and phone numbers are **masked first**, then Gemini picks **1 to 3 domains and a goal, only from the official lists**. Anything outside the list (like « Cuisine ») is removed by the server.
« Appliquer à mon parcours »	**Adds** the suggested domains (the others stay) and sets the goal.
« Enregistrer »	Saves the parcours. Nothing changes before this click.

4.3 Trainer → Studio → Apprenants

What you see	What happens behind
The 4 numbers (courses, learners, average progress, high-risk count)	Calculated from the enrolments in **this trainer's courses only** (a trainer never sees the learners of other trainers).
Table, « Suivi » column (Risque d'échec · 72 %…)	**Random Forest**, for each learner with complete data.
« Profil » column	**K-Means**.

What you see	What happens behind
Filters (Tous / Risque d'échec / À surveiller / Suivi normal, course, search)	No AI: they filter the table.
Click on a learner → side panel	Progress, time spent, last activity, and the risk box with **the model's 3 reasons**.
« Expliquer et proposer des actions » (LLM)	Gemini receives the risk, the reasons, the numbers and the missed notions. It writes a **careful summary** (« an estimate, not a certainty »), **2–3 actions for the trainer** and a **message to send to the learner**. The message must not mention risk, otherwise it is rejected.
« Copier le message »	Copies the suggested message so the trainer can send it.
« Générer le rapport » (LLM, class report)	The module calculates **group statistics only**: number at each risk level, average progress, profiles, most missed notions, per course. **No names.** Gemini writes a summary, **findings with numbers** and **actions**.

4.4 Admin → Supervision IA

What you see	What happens behind
« Données simulées » banner	The 3 models are **trained** on 200 simulated learners (a new platform has no history yet). The **predictions** use the real data.
The 4 scores (accuracy 82,5 %, recall 69 %, AUC 0,81, top 3 64 %)	Measured on **40 learners the model never saw** during training: an honest test.
Comparison chart	The Random Forest against 2 simpler methods, to prove it is better.
Confusion matrix	On 40 test learners: 22 correctly predicted as passing, 11 failures detected, 5 missed, 2 false alarms. Threshold 0,4 on purpose: **missing a learner in difficulty costs more than a false alarm**.
« Variables les plus importantes »	What weighs most in the risk: quiz average, correct answers, time spent…
K-Means curves (elbow, silhouette) and groups table	Why 3 groups is the best number, and what each group looks like.
Recommendation chart	Test: a followed course is hidden and the model must find it again. The hybrid finds it 64 % of the time in its top 3.
« Réentraîner »	Retrains the 3 models from scratch (about 1 minute). Same scores, because the simulated data uses a fixed seed: results are reproducible.
« Générer le rapport » (LLM, platform report)	Same as the class report, for **all the courses** of the platform.

5. The protections

1. **No personal data to the AI:** only numbers, course titles and notion names; any identifier sent is refused.
2. **Risk never shown to the learner**, even inside an LLM text: texts that mention it are rejected.
3. **The LLM cannot invent:** domains only from the list, plans within the declared time, numbers coming from the Machine Learning.
4. **Quota saved:** each text is generated on a click, saved, and reused while the data does not change.
5. **If Gemini is unavailable:** a clear message is shown, and all the Machine Learning results keep working.
6. **Roles:** a learner cannot see the risk or the Apprenants page; only the admin sees Supervision IA and can retrain.

6. In one sentence for the jury

« My module combines three classic Machine Learning models to understand each learner (profile, risk, recommendations) with a generative AI that turns those results into advice, plans and reports, without ever sending personal data and without ever showing the risk to the learner. »